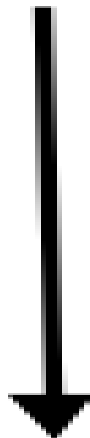


20/50 EMA Crossover Backtest: AAPL, GLD and DKN

How much did I make → how quickly → how much risk did I take → how bad did losses get → how often did my trades work → how does that compare with simply holding the stock?

Metric	What it means	Simple Qs it answers
Total return	The total % your investment gained or lost over the entire period	How much money did I make overall?
Annualised return (Derivation in Notes for Quant Book) - CAGR	Converts the total return into an equivalent compounded return per year	What did I earn per year on average? What constant % return every year compounded that turned x into y
Annualised Volatility (252 Trading Days)	Measures how much your daily returns fluctuate, scaled to a year	How much did my returns move around / how risky was it?
Sharpe ratio	Compares return earned with the amount of volatility/risk taken	Was the return good relative to the risk? Across all the individual daily observations in my backtest, what was my average excess return relative to the volatility of those returns?"
Maximum drawdown	The largest fall from a portfolio's previous peak to a later low	What was the worst decline I would have experienced?
Win Rate	Percentage of completed trades that made a profit	How often did my trades actually win? However win rate doesn't account for the size of wins and losses

Following pages are on each stock/assets



AAPL

	EMA Strategy	Buy-Hold Approach	Interpretation
Total return	88.7%	163%	Buy & Hold returned more
Annualised return	17.2%	27.4%	Buy & Hold returned more per year
Annualised Volatility	23.3%	33.6%	EMA experienced lower return variability
Sharpe ratio	0.627	0.771	Buy & Hold had higher return relative to volatility under your Sharpe assumptions
Maximum drawdown	20.4%	31.4%	EMA experienced a smaller worst peak-to-trough decline
Win Rate	55.6%	N/A	Just over half of completed EMA trades were profitable

AAPL — Analysis and Evaluation

The results show that the EMA strategy gave up some return in exchange for lower risk and market exposure. Buy & Hold generated a higher total and annualised return, but it also experienced higher annualised volatility and a larger maximum drawdown. This suggests that moving out of the market following bearish EMA crossovers reduced some downside exposure, but also meant that the strategy missed some of AAPL's upward price movements. In addition to this, despite the EMA strategy having lower volatility, Buy & Hold still achieved a higher Sharpe ratio, indicating that its additional return was enough to compensate for the additional risk taken.

The backtest assumes a zero return on cash while the strategy is out of the market. However, a more realistic extension could allow uninvested cash to earn a risk-free return, which is consistent with the risk-free rate used in the Sharpe ratio calculation. The model also excludes transaction costs and slippage, which would likely reduce strategy returns because each entry and exit introduces trading costs as well. Another possible extension would be a 200-day EMA trend filter, where long positions are only taken when the price is above its longer-term trend, potentially helping to avoid longer-term downtrends. However, this should be tested on new data to see whether it genuinely improves the strategy rather than just improving its performance on the historical AAPL data used to develop it.

GLD

	EMA Strategy	Buy-Hold Approach	Interpretation
Total return	5.78%	32.8%	Buy & Hold substantially outperformed the EMA strategy
Annualised return	1.42%	7.35%	Buy & Hold generated a considerably higher annual return
Annualised Volatility	12.1%	15.7%	EMA reduced return variability, but only moderately
Sharpe ratio	-0.153	0.277	EMA produced a negative risk-adjusted excess return under the 4% risk-free-rate assumption
Maximum drawdown	21.1%	22.0%	EMA provided very little reduction in the worst peak-to-trough decline
Win Rate	33.3%	N/A	Only one-third of completed EMA trades were profitable

GLD — Analysis and Evaluation with AAPL vs GLD

The results show that the 20/50 EMA crossover strategy performed worse on GLD than Buy & Hold. The EMA strategy generated a total return of only **5.78%** compared to the **32.8%** for Buy & Hold. Even though the strategy reduced annualised volatility by **3.6** percentage points, this reduction in risk came at the cost of a significant amount of return.

Unlike with AAPL, the EMA strategy provided little to no improvement in maximum drawdown, with a maximum decline of **21.1%** compared with **22.0%** for Buy & Hold. The equity curve also shows many periods where the EMA strategy remained out of the market following bearish crossovers. This helped reduce exposure, but also meant that the strategy failed to capitalise on some GLD price increases. The EMA strategy also produced a negative Sharpe ratio of **-0.153** with the assumed 4% annual risk-free rate, which indicates that, after accounting for the assumed risk-free return, the strategy does not generate enough return to compensate for the volatility it experienced.

The difference between the AAPL and GLD results can be explained by the different market conditions experienced by each asset during the testing period, as the 20/50 EMA crossover is a trend-following strategy and therefore relies on a persistent directional price movement. GLD experienced several periods of more directionless or oscillating price behaviour, particularly during late 2020–2021 and parts of 2022. During these periods, the EMA20 repeatedly moved around the EMA50, generating false signals and whipsaws as the strategy entered positions only for the price direction to reverse soon afterwards.

We can see whipsaws occur from October to November 2020 on the backtest chart, causing repeated whipsaws as the strategy entered a position only for the price direction to reverse soon afterwards. This helps explain the lower **33.3%** win rate and much weaker overall return for GLD, as well as the significant effect on our strategy's equity curve, where it

declined and then remained relatively flat for an extended period. An example of a false signal occurred around early January 2021, when a bullish EMA crossover generated a BUY signal but the upward movement failed to persist. GLD reversed direction, and the strategy later exited the position. This demonstrates an important weakness of moving-average crossover strategies: because the indicators are based on past prices (making them lagging indicators), only for that movement to reverse shortly afterwards.

In contrast, AAPL experienced stronger and more sustained upward trends during the sample, particularly through parts of 2020–2021 and again during 2023. These directional movements allowed the EMA20 to remain above the EMA50 for longer periods, meaning the strategy remained invested and was able to benefit from larger price movements. This helps explain why the strategy achieved a **55.6%** win rate on AAPL compared with only **33.3%** on GLD.

These results are particularly useful because the same 20/50 EMA rules and 2020–2023 testing period were applied to both assets without changing the parameters. The strategy reduced volatility for both assets, but its effectiveness was not consistent. On AAPL, it achieved an **88.7%** return with a **20.4%** maximum drawdown, whereas on GLD it achieved only a **5.78%** return while still experiencing a **21.1%** maximum drawdown. This suggests that the effectiveness of a trend-following EMA crossover depends significantly on the price behaviour and market regime of the underlying asset. Lastly, the same limitations mentioned for AAPL apply to both backtests.

Extension (DKNG)

	EMA Strategy	Buy-Hold Approach	Interpretation
Total return	310.1%	230.1%	EMA outperformed Buy & Hold
Annualised return	42.3%	34.8%	EMA generated a higher compounded annual return
Annualised Volatility	51.8%	73.4%	EMA experienced lower return variability
Sharpe ratio	0.863	0.723	EMA generated better return relative to volatility under the Sharpe assumptions
Maximum drawdown	48.4%	85.7%	EMA reduced the worst peak-to-trough decline
Win Rate	66.7%	N/A	Two-thirds of completed EMA trades were profitable

Background — While watching the 2026 World Cup and learning the basics of this project, I became interested in whether major external events could affect the performance of the strategy, either creating opportunities or causing it to break down. I therefore decided to test the strategy on DraftKings (DKNG), a sports betting company, and investigate whether there was any noticeable change in its price behaviour or EMA signals around the 2022 World Cup. However this became my best strongest result for a completely different reason.

DKNG — Analysis and Evaluation

The 20/50 EMA crossover strategy performed strongly on DKNG over the testing period and, unlike the AAPL and GLD results, outperformed Buy & Hold. The EMA strategy generated a total return of **310.1%**, compared with **230.1%** for Buy & Hold. It also achieved a higher annualised return of **42.3%**, compared with **34.8%**, showing that the strategy produced stronger compounded returns over the four years. Importantly, this higher return was achieved with lower volatility. The EMA strategy had annualised volatility of **51.8%**, compared with **73.4%** for Buy & Hold. This is also reflected in the Sharpe ratios: with the EMA strategy achieving a Sharpe ratio of **0.863**, compared with **0.723** for Buy & Hold. Under the assumed 4% risk-free rate, the strategy therefore generated a better return relative to the volatility experienced. The strategy also achieved a **66.7% win rate**, meaning two-thirds of its completed trades were profitable. This was considerably higher than the 55.6% win rate on AAPL and 33.3% on GLD. However, win rate alone does not explain the strong performance, as it does not account for the size of individual gains and losses. Furthermore, Buy & Hold experienced a maximum drawdown of **85.7%**, compared with **48.4%** for the EMA strategy.

DKNG experienced a large and persistent decline from late 2021 through much of 2022. Once the EMA20 moved below the EMA50, the long-only strategy moved out of the market and remained in cash for much of this downward trend. The equity curve shows the effect clearly: while the value of Buy & Hold fell substantially, the EMA strategy remained relatively flat while it was out of the market. Eventhough the strategy did not profit from the falling price, instead, it avoided participating in a significant portion of the decline.

GLD experienced more directionless and oscillating periods, which caused repeated crossovers, false signals and whipsaws. DKNG experienced much larger and more persistent directional movements. Although DKNG itself was considerably more volatile, the EMA crossover was more effective at identifying periods when the strategy should be invested or out of the market. This demonstrates that the success of the 20/50 EMA strategy depends not simply on the overall volatility of an asset, but on whether its price movements develop into sufficiently persistent trends.

DKNG — World Cup Effect

My initial reason for selecting DKNG was to investigate whether a major external sporting event such as the 2022 FIFA World Cup could affect the price behaviour of a sports betting company and influence the EMA crossover strategy. My initial assumption was that increased attention and betting activity surrounding the tournament might create a market movement and be captured by the strategy.

However, the backtest doesn't provide evidence supporting this hypothesis. During the World Cup period from 20 November to 18 December 2022, the EMA20 remained below the EMA50 and the strategy therefore remained out of the market. No bullish EMA crossover was generated during the tournament. The World Cup may have influenced factors such as customer activity or betting volumes, but this backtest alone cannot establish whether it caused changes in DKNG's share price. Based on the EMA signals, the event did not create a trading opportunity under the 20/50 crossover rules.